CYC475 - Topics in Cyber Security (Web Technologies and Programming)

WEEK 1 & 2: HTML

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Lecture Contents

- HTML: Introduction, History, HTML and XHTML,
- HTML Elements,
- Block Level vs Inline Elements,
- HTML Lists and Tables
- HTML div and span
- HTML Validator.
- HTML layout design
- HTML Semantic elements
- HTML 5 elements

HTML

- HTML
  - Stands for hypertext markup language
  - Hypertext: text that has reference to other text
- HTML is not a programming language. It is a markup language.
- A markup language is a text-encoding system which specifies the structure and formatting of a document and potentially the relationships among its parts.
- A markup language is both machine and human readable

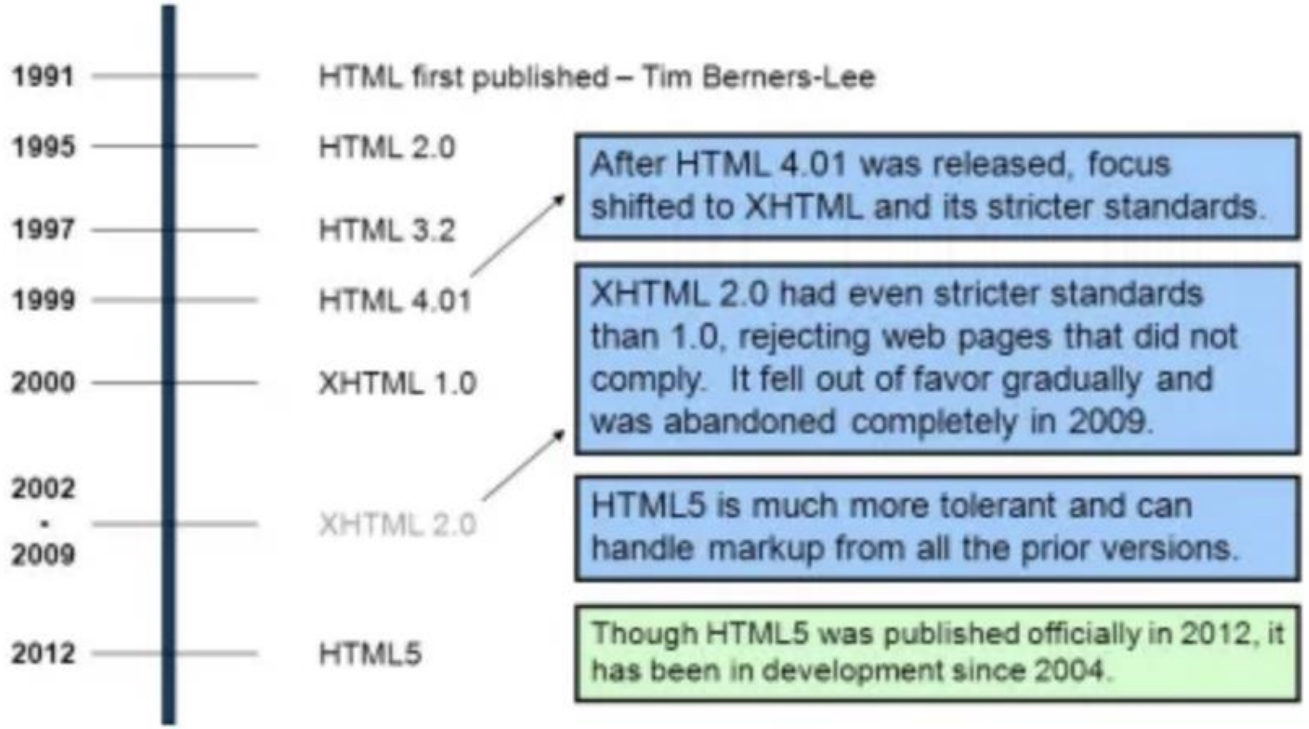
HTML history

HTML was created by Berners Lee in late 1991.

HTML 2.0 was the first standard HTML specification which was released in 1995.

XHTML was developed to make HTML more extensible and flexible to work with other data formats (such as XML). In addition, browsers ignore errors in HTML pages, and try to display the website even if it has some errors in the markup. So XHTML comes with a much stricter error handling.

HTML is no longer released as big, numbered versions (like HTML4, HTML5). Instead, it's now a **"living standard"** maintained by **WHATWG**.



Three layers of web design:

Structure	Style	Behavior
HTML markup	CSS	JavaScript

Text + HTML + CSS = Web Page

A simple HTML document

- The <!DOCTYPE html> declaration defines that this document is an HTML5 document
- The <html> element is the root element of an HTML page
- The <head> element contains meta information about the HTML page
- The <title> element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph

```html
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

```
<html>
    <head>
        <title>Page title</title>
    </head>
    <body>
        <h1>This is a heading</h1>
        <p>This is a paragraph.</p>
        <p>This is another paragraph.</p>
    </body>
</html>
```

The <!DOCTYPE> Declaration

- The <!DOCTYPE> declaration represents the document type, and helps browsers to display web pages correctly.
- It must only appear once, at the top of the page (before any HTML tags).
- The <!DOCTYPE> declaration is not case sensitive.
- The <!DOCTYPE> declaration for HTML5 is:

```
<!DOCTYPE html>
```

Even though HTML has moved beyond HTML5, we still write the above standard. It's the standard doctype for the **living HTML specification**.

HTML Element

- An HTML element is defined by a start tag, some content, and an end tag:
- <tagname> Content goes here... </tagname>
- The HTML **element** is everything from the start tag to the end tag:
- <h1>My First Heading</h1>
- My first paragraph.
- Some HTML elements have no content (like the <br> element). These elements are called empty elements. Empty elements do not have an end tag!
- Never skip the end tag, though it may not cause an error.
- HTML is not case sensitive and <P> are the same.

Attributes

- Attributes Tell Us More About Elements
  - Attributes provide additional information about the contents of an element.
  - They appear on the opening tag of the element
  - and are made up of two parts: a name and a value, separated by an equals sign.

Attributes

ATTRIBUTE NAME

```
<p lang="en-us">Paragraph in English</p>
```

ATTRIBUTE VALUE

The attribute **name** indicates what kind of extra information you are supplying about the element's content. It should be written in lowercase.

The **value** is the information or setting for the attribute. It should be placed in double quotes. Different attributes can have different values.

Here an attribute called `lang` is used to indicate the language used in this element. The value of this attribute on this page specifies it is in US English.

Title

```
<html>
  <head>
    <title>This is th
  </head>
  <body>
    <h1>This is the Body of the Page</h1>
    <p>Anything within the body of a web page is
    displayed in the main browser window.</p>
  </body>
</html>
```

This contains information that are shown on top of the page

Body & Title

This is the Title of the Page
www.CUSTSemester6.com/example2.html
This is the Body of the Page
Anything within the body of a web page is displayed in the main browser window.

Headings

- HTML has six "levels" of headings:
  - **<h1>** is used for main headings
  - **<h2>** is used for subheadings
  - If there are further sections under the subheadings then the **<h3>** element is used, and so on...

Paragraph

- A paragraph consists of one or more sentences that form a kind of dialogue.

```
<html>
<head>
<title>This is the Title of the Page</title>
</head>
<body>
<p>this is a paragraph and
when you press enter, does it moves to
new line?</p>
<p>yet another paragraph</p>
</body>
</html>
```

Bold and Italics

- Bold
  - By enclosing words in the tags **** and **** we can make characters appear bold.
- Italic
  - By enclosing words in the tags **** and **** we can make characters appear italic.
- Example

```
<html>
<head>
<title>This is the Title of the Page</title>
</head>
<body>
<b>Does it appears bold?</b>
<i>and it must be italic</i>
<b><p> bold but in a new paragraph</p></b>
</body>
</html>
```

Line Break

- Line Break
  - Just like the line break <br /> breaks line and starts new line
  - Example

```
<p>The Earth<br />gets one hundred tons heavier
every day<br />due to falling space dust.</p>
```

- Horizontal Rule
  - Just like <br /> it separates lines by placing a solid line in between paragraphs

```
<p>Venus is the only planet that rotates
clockwise.</p>
<hr />
<p>Jupiter is bigger than all the other planets
combined.</p>
```

Blockquote

- Blockquote
  - The <blockquote> tag specifies a section that is quoted from another source.
  - Browsers usually indent <blockquote> elements.

```html
<!DOCTYPE html>
<html>
  <body>
    <h1>The blockquote element</h1>
    <p>Here is a quote from WWF's website:</p>
    <blockquote cite="http://www.worldwildlife.org/who/index.html">
    For 50 years, WWF has been protecting the future of nature....
    </blockquote>
  </body>
</html>
```

The blockquote element

Here is a quote from WWF's website:

> For 50 years, WWF has been protecting the future of nature....

Block level and inline elements

- Every HTML element has a default display value, depending on what type of element it is.
- The two most common display values are block and inline.

Block-level elements

- A block-level element always starts on a new line, and the browsers automatically add some space (a margin) before and after the element.
- A block-level element always takes up the full width available (stretches out to the left and right as far as it can).
- Two commonly used block elements are: <p> and .
- The element defines a division or a section in an HTML document.

```
<div style="background-color:black;color:white;padding:20px;">
  <h2>London</h2>
  <p>London is the capital city of England. It is the most populous city
in the United Kingdom, with a metropolitan area of over 13 million
inhabitants.</p>
</div>
```

Block-level elements

Below is a list of all the block-level elements:

```
<address>   <article>   <aside>     <blockquote> <canvas>    <dd>        <div>
<dl>        <dt>        <fieldset>  <figcaption> <figure>    <footer>    <form>
<h1>–<h6>   <header>    <hr>        <li>         <main>      <nav>       <noscript>
<ol>        <p>         <pre>       <section>    <table>     <tfoot>     <ul>
<video>
```

Inline elements

- An inline element does not start on a new line.
- An inline element only takes up as much width as necessary.
- This is a element inside a paragraph.

```
<p>My mother has <span style="color:blue;font-
weight:bold;">blue</span> eyes and my father
has <span style="color:darkolivegreen;font-weight:bold;">dark
green</span> eyes.</p>
```

Inline elements

<a>	<abbr>	<acronym>	<b>	<bdo>	<big>	
<button>	<cite>	<code>	<dfn>		<i>	<img>
<input>	<kbd>	<label>	<map>	<object>	<output>	<q>
<samp>	<script>	<select>	<small>			<sub>
<sup>	<textarea>	<time>	<tt>	<var>		

Lists

- There are two kinds of lists in HTML
- Ordered list: with <ol> tag and <li> tags for list items
- Unordered list: with <ul> tag and <li> tags for list items

```html
<ul>
    <li>list item 1</li>
    <li>list item 2</li>
    <li>list item 3</li>
</ul>
<ol>
    <li >list item 1</li>
    <li>list item 2</li>
    <li>list item 3</li>
</ol>
```

Lists

- Definition lists:
- HTML definition lists is used to define a description list, it is represented by using definition list container **<dl>** tag.
- It is used along with definition term **<dt>** and definition description **<dd>**.
- In HTML Description list or definition list displays its elements in definition form in the dictionary where if we define a description list it will give a description of each item in the list by using the following tags.

Tables

```
<table>
    <tr>
        <th>Company</th>
        <th>Contact</th>
        <th>Country</th>
    </tr>
    <tr>
        <td>Comsats</td>
        <td>CS department</td>
        <td>Pakistan</td>
    </tr>
</table>
```

The W3C Markup validation service

- The Markup Validation Service is a validator by the World Wide Web Consortium that allows Internet users to check pre-HTML5 HTML and XHTML documents for well-formed markup against a document type definition.
- Markup validation is an important step towards ensuring the technical quality of web pages.
- https://validator.w3.org/detailed.html

HTML element

- The element is used as a container for other HTML elements.
- The element is by default a block element, meaning that it takes all available width, and comes with line breaks before and after.

HTML element

- The tag is an inline container used to mark up a part of a text, or a part of a document.
- The tag is easily styled by CSS or manipulated with JavaScript using the class or id attribute.
- The tag is much like the element, but is a block-level element and is an inline element.

HTML Layout Design

- UI designers normally make website designs in some tools like Figma.
- Translating that design into a web page is the job of a front end developer.
- He/she needs to identify the layout, sections, and various css properties involved in that design.
- Have a look at some websites and see how you can design them.

Activity

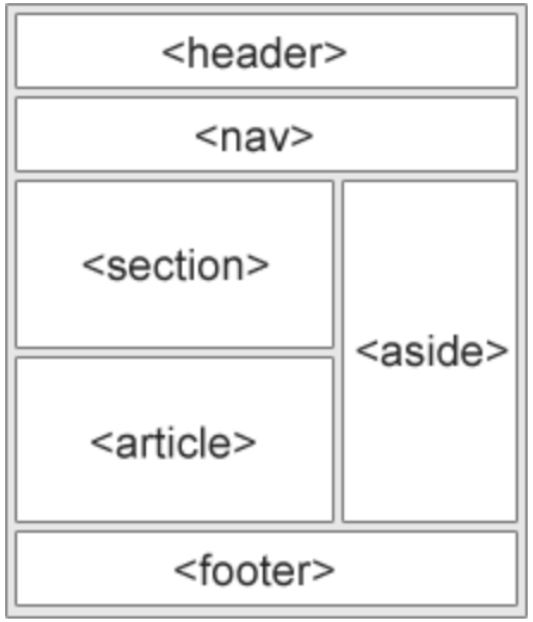
- Create a website that gives information about your course. The website should have the following navigation menu:
  - Home
  - Course Details
  - Instructor details
  - Students Details
- Think about the layout of your website and prepare a wireframe design on a page. Your website should include:
- Headings, paragraphs, links, images, lists and tables with navigation. You can link Comsats website and other social media links in the footer.

HTML Semantics

- A semantic element clearly describes its meaning to both the browser and the developer.
- Examples of **non-semantic** elements: and - Tells nothing about its content.
- Examples of **semantic** elements: <form>, <table>, and <article> - Clearly defines its content.

HTML Semantics

Why Semantic Elements?
According to the W3C: "A semantic Web allows data to be shared and reused across applications, enterprises, and communities."



HTML 5

- Semantic Web
  - In previous versions of HTML common structural elements such as navigation menus, page headers, and main content sections were not semantically differentiated. Instead, they all used the tag.
  - HTML5 addresses this with a host of new semantic elements, such as <header>, <main>, <section>, <nav>, <aside>, <article>, and <footer>.
  - Additionally, new inline semantic elements such as <address> and <time> help online services such as search engines quickly locate relevant data on a page.

HTML 5

- Rich media experiences without plugins:
  - HTML5 inherently supports rich media through elements such as <video> and <audio>.
  - Some disadvantages of plugins such as Java and Flash include poor performance, fewer user options, security flaws, and lack of search engine visibility.
- XML compatibility:
  - XHTML5 code can be written using the ‘stricter’ XML syntax.
  - This is useful for developers who prefer the neatness offered by well-formed XHTML, including quoted attribute values, lowercase element names, and the closure of all elements.
  - In cases where code is expected to work with other XML applications, HTML5 documents must be served as XML.

HTML 5

- Design & content separation:
  - HTML5 discourages **non-meaningful markup** that is only intended to help the browser visualize stuff (think declarations for font, text color, text alignment, and so on).
  - This standard has deprecated many elements that enabled such visualization, and the few supported features display '**not a recommended practice**' warnings.

HTML 5

- Accessibility and design responsiveness:
  - Early iterations of HTML were *not seamlessly compatible with modern technology's numerous screen sizes and aspect ratios*.
  - This includes mobile phones, tablets, e-readers, assistive technologies such as text-to-speech converters, screen readers that suppress styling while boosting text magnification and contrast, and braille interpreters.
  - These technologies were held back by markup that attempted to 'hard-code' styling and design into page content

HTML 5

- APIs:
  - HTML5 has made a great leap by defining numerous new APIs that simplify communication with applications.
  - It introduces APIs for functionality that earlier needed plugins or customized code, including web storage, drag and drop, geolocation, and microdata.

HTML Quotation and Citation Elements

Tag	Description
<abbr>	Defines an abbreviation or acronym
<address>	Defines contact information for the author/owner of a document
<bdo>	Defines the text direction
<blockquote>	Defines a section that is quoted from another source
<cite>	Defines the title of a work
<q>	Defines a short inline quotation

HTML Abbreviation

- The HTML <abbr> tag defines an abbreviation or an acronym, like "HTML", "CSS", "Mr.", "Dr.", "ASAP", "ATM".
- Marking abbreviations can give useful information to browsers, translation systems and search-engines.

```
<!-- Abbreviation -->
<p>The <abbr title="World Health Organization">WHO</abbr> was founded in 1948.</p>
```

HTML Entities

Result	Description	Name	Number
	non-breaking space		
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark	'	'
¢	cent	¢	¢
£	pound	£	£
¥	yen	¥	¥
€	euro	€	€
©	copyright	©	©
®	trademark	®	®

Activity

- Identify the divisions/tags in your previously designed website that can be converted to semantic tags. Redesign using semantic tags.